

Professional Summary

Software Engineer with 3+ years of experience progressing from Python/SQL application development into backend services, cloud-connected workflows, and AI-enabled software. Strong in Python, Java, C++, SQL, REST APIs, debugging, performance optimization, testing, and cloud deployment. Delivered production workflows with 99.8% uptime, 40% throughput gains, and 60% faster distributed processing.

Technical Skills

Languages	Python, Java, C++, SQL, JavaScript, TypeScript, R, Bash
Backend & APIs	Flask, FastAPI, REST APIs, API Design, Microservices, JSON, HTTP, Backend Services
Engineering	Object-Oriented Programming, Data Structures & Algorithms, Debugging, Performance Optimization, Unit Testing, Integration Testing, Pytest, Git, GitHub, Docker, Linux, CI/CD
Databases & Data	PostgreSQL, MySQL, Microsoft SQL Server, Snowflake, Google BigQuery, Database Design, ETL/ELT, Data Modeling, Query Optimization
AI & ML	LLM Integration, RAG, OpenAI API, Prompt Engineering, LangChain, LlamaIndex, Generative AI, NLP, Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, XGBoost, MLflow, PySpark
Cloud & Production	AWS (S3, EC2, Lambda, CloudWatch), Azure, Airflow, Monitoring, Logging, Production Testing

Experience

Graduate Student Assistant — AI-Enabled Software Systems Jan 2026 – Present

University at Buffalo — UB Media Forensic Lab — Advisor: Prof. Siwei Lyu, IEEE Fellow

- Built Python data-processing components for 50K+ multimodal records in AWS S3, cutting preprocessing errors by 45% for the Deep-O-Meter research platform and improving downstream data readiness.
- Tuned distributed PySpark workloads for multimodal ML processing, reducing computation time by 60% through efficient partitioning and joins while supporting large-scale feature processing.
- Exposed Flask REST APIs for deep-learning inference, lowering latency from 580ms to 378ms and increasing throughput by 28%.
- Evaluated 5+ deep-learning architectures through standardized validation workflows, raising synthetic-media classification F1-score by 12%.
- Streamlined MLflow experiment tracking across 10+ configurations, reducing validation effort by 30%.

Senior Software Engineer — Backend, Cloud & Integration Systems

Sep 2024 – Jul 2025

Broadridge Financial Solutions, India

- Automated Python workflows connecting AWS S3, SFTP, and enterprise systems; processed 200+ files daily, increasing throughput by 40% and eliminating 15 hrs/week of manual processing.
- Integrated Sterling File Gateway with 5+ enterprise systems, adding validation, exception routing, and reliable file-handling paths that lowered cross-system errors by 35%.
- Established monitoring and recovery with CloudWatch, PagerDuty, and Airflow across 12+ production workflows, maintaining 99.8% uptime and reducing incident resolution time by 25%.
- Refactored 20+ SQL queries using CTEs and window functions and redesigned 4 schemas, decreasing query latency by 30% and storage overhead by 20%.
- Instituted regression tests and deployment checks across 15+ workflows, cutting post-release defects by 40%.

Software Engineer — Python Applications & Data Systems

Aug 2022 – Aug 2024

Freelance, Remote

- Created Python/SQL applications handling 500K+ records across 10+ data sources, accelerating execution by 45% through vectorized transformations.
- Restructured relational schemas and indexing strategies for 8 datasets, achieving 60% faster queries across 3 enterprise clients.
- Delivered 15+ client applications and reporting interfaces using Python and SQL, translating client requirements into reusable software components and data-driven interfaces.
- Strengthened application workflows with validation and error recovery, maintaining 99.5% accuracy across 2M+ records.
- Applied Python and SQL debugging and performance tuning across client workflows, strengthening reliability and maintainability of delivered applications.
- Applied Python and SQL debugging, performance tuning, and application-level validation across client workflows, improving reliability and maintainability of delivered software.

Projects

- Real-Time Streaming Platform (Kafka, Spark Streaming, AWS S3):** Streamed 500K+ events/hour with sub-2s latency; sustained 99.9% uptime during 3× traffic spikes.
- Distributed Data Processing (PySpark, Hadoop):** Transformed 100GB+ datasets, reducing batch runtime by 65% through broadcast joins and adaptive partitioning.
- Workflow Orchestration Platform (Airflow, PostgreSQL):** Authored 10+ DAGs for multi-step ETL workflows, eliminating 20+ hrs/week of manual scheduling and reaching 98% SLA compliance.
- Azure Data Warehouse (ADF, Snowflake, SQL):** Unified 6 sources and transformed 50M+ records/month, reducing load SLA from 4 hrs to 45 min.

Education

University at Buffalo — M.S. Data Science

Aug 2025 – Dec 2026

Coursework: Machine Learning, Statistical Learning, Data Mining, Predictive Analytics, Database Systems, Numerical Computing

Jawaharlal Nehru Technological University Hyderabad (JNTUH) — B.Tech. Computer Science

Aug 2019 – Jul 2023

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning, Cloud Computing, Distributed Systems

Certifications

Cloud: AWS Essentials for Developers · Azure Fundamentals (AZ-900) · Azure AI Fundamentals (AI-900)

Data & Analytics: IBM Data Science Professional · IBM Data Analyst Professional · IBM Data Visualization Core

Generative AI: Microsoft & LinkedIn Career Essentials in Generative AI Foundation